

Software Testing Fundamentals

Regression testing is a type of software testing that verifies that previously working functionality still works after changes are made to the codebase. It is performed after bug fixes, feature additions, or refactoring to ensure no new issues have been introduced.

Unit testing focuses on testing individual components or functions in isolation. A unit test verifies that a single piece of code works as expected, independent of other parts of the system. Unit tests are fast to run and easy to debug.

Integration testing checks that multiple components work correctly when combined. It bridges the gap between unit tests and end-to-end tests by verifying the interactions between modules, services, and external dependencies.

Test coverage is a metric used to measure the percentage of source code executed during a test run. High test coverage generally indicates a more thoroughly tested codebase, though 100% coverage does not guarantee the absence of bugs.

The testing pyramid is a concept that describes the ideal distribution of different test types: many unit tests at the base, fewer integration tests in the middle, and even fewer end-to-end tests at the top. This structure helps maintain a fast, reliable, and maintainable test suite.

Exploratory testing is an informal, simultaneous approach to learning, test design, and test execution. Unlike scripted testing, exploratory testing relies on the tester's skill and intuition to discover unexpected defects.